

Brief report of New Booster Pump-01 seal failure at WRPL Viramgam under SMPL DBL Project

Introduction-

Total 5 nos. of new booster pumps have been commissioned under SMPL DBN project on second week of January 2017. BP-01 & BP-02 commissioned on 12.01.2017 and MP-03, BP-04 & BP-05 commissioned on 13.01.2017.

Name plate Details of these 5 nos. is given below-

S.N.	Pump	Rated power	Rated Head	Rated Flow
1.	NBP-01,02 &03	605 KW	50 m	2100 KL/hr
2.	NBP-04&05	380 KW	50 m	1325 KL/hr

Brief Description of Breakdown-

1. On 01.03.2017, seal failure on BP-01 has been reported after cumulative running of 160 hrs when pump was not running. After dismantling, stationary part of seal silicon carbide got completely damaged.
2. During commission under DBN project, Total 2 nos. of complete cartridge type mechanical seal for BP-1 & BP-2 received and used in BP-02 breakdown occurred earlier on 17.02.2017. Therefore the mechanical seal spares for higher capacity pump (BP1 & BP2) was nil and BP-01 was not-available till 17.03.2107. However simultaneously local seal was fabricated by M/s Apex. The same was fitted and trial run taken on 18.03.2017 but after running of only 9 hrs , the seal was again failed and found completely broken as shown in figure-1.
3. Pump was again not available till 06.04.2017. However instead of silicon carbide, tungsten carbide seal was fabricated and fitted on 06.04.2017 but again seal failed after installation. Pump is not available since then.



Fig-01 : Carbon completely broken



Fig-02 : Seal failure

Maintenance activities done after BP-01 Breakdown-

1. Alignment has been checked and found 0-0.
2. Vibration has been checked and found maximum value of 1.6 mm/s on NDE motor horizontal.
3. Run out in the shaft measured which is 260 microns.
4. Completion of shelf life of spares (OES suggested 2 years of shelf life).

Since, frequent failures of seals were experienced in the pumps it was decided to dismantle the pump and get it inspected.

Maintenance Activities carried out at Viramgam station for dismantling the NBP-01-

1. Pump and motor assembly dismantled.
2. Mechanical seal dismantled.
3. Complete pump shaft assembly was not able to lift as height of pump is 19 Ft (5.8 m) which is more than the effective lift (4.6 m of the EOT crane).
4. It was decided to dismantle the pump shaft in parts.
5. Clamps are locally fabricated to hold the part shaft.
6. Pump shaft was dismantled in three parts.
7. Pump shaft with 2 stage impeller along with part shafts and spares sent to M/s Allwyn Industries for further maintenance on 07.06.2017.



Fig-03 : Dismantling of Booster at Site



Fig-04 : Main shaft removal

Maintenance Activities carried out at Allwyn workshop –

- Dismantled Pump having 3 intermediate shafts, main impeller shaft & header shaft along with couplings and spares received at Allwyn Industries for further maintenance on 07.06.2017.
- Maintenance activities carried out at Allwyn is given in the following table.

S.N.	Service description	Maintenance description
1.	<u>Cleaning</u>	Cleaning of Booster pump done thoroughly
2.	<u>Ultrasonic NDT testing</u>	Ultrasonic NDT testing of all shafts, shaft sleeves, wear rings done and found OK.
3.	<u>Run-out check</u> a) <u>Header shaft- 01 no.</u> b) <u>Intermediate- 03 nos.</u> c) <u>Main shaft- 01 no.</u>	Run-out of all pump shafts checked and measured. a) Run-out of header shaft is within limits of 0.07 mm , only polishing done. b) Run out found in all 3 intermediate shaft along with heavily scouring marking so machining, metallizing and polishing done to remove run out and scouring present in shafts. c) Run out of main shaft is found to be 0.1 mm along with scouring marking so shaft straightening , machining , metallizing and polishing done to remove run out d) Therefore polishing was done in all shafts to remove run-out within limits.
4.	<u>Muff Couplings</u>	Run out of all 4 Muff couplings found to be from 0.45 mm to 0.65 mm. Run-out in muff coupling was one of the probable reason of seal leakage as it acts a power transmitting media between the shafts. Therefore repairing done to remove run-out and further inspection of muff coupling was carried out on lathe machine and found OK.
5.	<u>Checking of shafts</u>	Scoring marking & uneven marking was also observed in different segments of shaft. Polishing done.
6.	<u>Spider Bush</u> a) <u>Spool part</u> b) <u>Pump lower part</u>	Spider bush was found damaged (concentric marking on inner side- Photograph attached) a) 3 Nos. of spider bush of spool found excessive scouring marks so these bushes are newly fabricated and fitted . Clearance between sleeve and bush was kept in the range of 0.17 mm to 0.22 mm. b) Light scouring marks found on 3 nos. of spider bush in lower part of pump therefore boring and machining of spider bush done and simultaneously metallizing of shaft done.
7.	<u>Sleeve repairing</u>	Total 3 nos. of sleeve found scouring marks and machined.
8.	<u>Wear Rings</u>	Scoring markings of case wear ring & impeller wear rings observed and machined.
9.	<u>Spool parts</u>	Face run out found. Spool no. 1, 2 & 3 found face run out of 0.2 mm , 0.1 mm & 0.1 mm respectively. In all spool pieces, distortion of bush bearing journal found. Therefore New bush fitted with liquid nitrogen with interference fit.
10.	<u>Dynamic Balancing</u>	Dynamic balancing of pump rotor assembly, shaft with hub and coupling done after repairing and found within limits
11.	<u>UT Test</u>	Ultrasonic NDT testing of re-metallized shafts, shaft sleeves, wear rings done.

- Pump assembly was done in two parts on 10.07.2017. First part is Main shaft with two spool pieces. 2nd part is Discharge barrel with 1 no. of spool piece and bearing housing.
- Complete booster pump was sent to Viramgam from Allwyn industries on 16.07.2017 and received on 17.07.2017.



Fig-05 : Scouring marks on shaft



Fig-06 : Marking on Spider Bush



Fig-07 : Pump rotor assembly after repair



Fig-08 : Muff Coupling run out checking

- Budgetary quotation by M/s Allwyn Industries was provided. Estimated cost for repairing job of NBP-01 is Rs. 5.5 Lakh.



Fig-09 : Pump rotor assembly before repairing



Fig-10 : Pump assembly after maintenance

- Maintenance activities started from 17.07.2017 at Viramgam station. Pump having main impeller shaft casing, 3 intermediate shafts spool pieces & header shaft housing fitted till 22.07.2017.
- Installation of Booster pump after repairing got delayed due to adverse weather condition at Viramgam.
- After installation of motor, mechanical seal (Make- Eagle Burgmann) fitted. Load trial of NBP-1 taken and all parameters found normal..